INCH-POUND

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SUPERSEDING
MIL-M-70453 (AR)
1 April 1985

MILITARY SPECIFICATION

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for one type of mount known as Mount, Telescope: 9360169 used with the U.S. Army mortar systems (see 6.1). Throughout this specification this instrument is referenced as "mount." This mount has radioactively illuminated scales, level vials, and indices (see 3.2.1 & 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-A-225/6 - Aluminum Alloy 2024, Bar, Rod and wire, Rolled, Drawn or Cold Finished

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1240

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QQ-A-225/8 - Aluminum Alloy 6061, Bar, Rod, Wire & special Shapes, Rolled, Drawn or Cold Finished

MILITARY

MIL-F-13926 - Fire Control Material: General Specification Governing the Manufacture and Inspection of
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-T-70521 - Telescope, Elbow: 9360257

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents. drawings. and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.4)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

6523553 - Elevation and Cross Leveling Fixture
7680631 - Dioptometer
7691596 - Azimuth Test Fixture
8658940 - Inspection Aid
9360169 - Mount, Telescope
9360178 - Knob Assembly
9360180 - Scale Assembly
9360182 - Knob Assembly

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9360184 - Holder Assembly
9360257 - Telescope, Elbow
11741740 - Shaft Assembly
11741741 - Shaft Assembly
11741770 - Housing
11741774 - Gear
11741792 - Bracket Assembly
11750708 - Adapter, Final Inspection, Elevation
11750709 - Adapter, Final Inspection, Azimuth
11750712 - Alignment Gage, Final Inspection, Azimuth

DEPARTMENT OF DEFENSE, DEFENSE SUPPLY AGENCY

DLAM 4145.8 - Radioactive commodities in the DOD Supply System

PACKAGING DATA SHEET

PS 9360169 - Packaging of Mount, Telescope: 9360169

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following documrent(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

INDUSTRIAL

ASTM B211 - A11oY, Aluminum, Bar, Rod and Wire

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187).

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9360169, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the mount shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. workmanship.

3.2.1 Radioactive material Prospective bidders should be cognizant of the need for a Nuclear Regulatory Commission License governing the procurement possessing, handling, and shipping of radioactive material.

3.3 Environmental. Any mount subjected to any or all of the requirements of 3.3.1, 3.3.2, 3.3.3.1, and 3.3.3.2 inclusive, shall subsequently meet the requirements of 3.5 to 3.7.5 inclusive, unless otherwise specified.

3.3.1 vibration. The mount shall be vibrated at 30 $\pm$ 5 cycles per second frequency with an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of five minutes $\pm$ 15 seconds. Orientation of the mount shall be in accordance with

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requirement 3.6. In addition, the azimuth and elevation locking knobs should be turned to a locked position with a ten (10) inch-pound torque each.

3.3.2 Shock. The mount shall be shocked six (6) times in each of three (3) positions. The elevation scale shall be set at the 800 mil mark and the azimuth and elevation lockingknobs turned to a locked position with a ten (10) inch-pound torque each. The applied shocks shall be in a vertical, up direction In the first position, the mounting dovetail shall be in the horizontal plane and subjected to six (6) shocks, each at 300g for one-half millisecond. In the second and third positions, the mounting dovetail shall be in a vertical plane with its axis at 45° and 225° and subjected to six (6) shocks each at 200g for one millisecond. The shock pulse shall be one-half sine wave.

3.3.3 Temperature.

3.3.3.1 Operating. The mount shall meet the requirements of 3.7 and 3.8 inclusive, while exposed to ant thermally stabilized at ambient temperatures of -50°F and +145°F (-46°C and +63°C).

3.3.3.2 Non-operating. The mount shall show no evidence of physical failure such as adhesive separations, glass vial breakage, or casting failure after having been exposed and thermally stabilized at -50°F and +160°F (-46°C and +71°C).

3.4 External tritium contaminatiOn. Tritium in any form shall not be present on the external surfaces of the mount in excess of 1000 dpm/100 sq. cm. subsequent to testing for each of the following requirements: 3.3.1, 3.3.2, 3.3.3.1, and 3.3.3.2.

3.5 Illumination. the markings of the scales, indices, and level vials should be clearly distinguishable when observed with and without a 50% neutral density filter at light levels of zero to ten foot-candles at a distance of 18 to 20 inches from the viewers eye.

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3.6 Orientation. The mount shall be positioned as follows to meet the requirements of 3.7 and 3.8. Where line of sight is specified. the requirement shall be met with the line of sight entering the telescope mounting sleeve from the end opposite to the telescope elevation mechanism. Telescope, 9360257, (MIL-T-70521) or equivalent dummy load with a suitable adapter, shall be secured in the dovetail of the bracket assembly by the telescope clamping mechanism.

a. The mounting dovetail surface in a vertical plane.

b. The axis of the dovetail vertical.

c. A line of sight provided which is parallel to the above mounting surface and which is perpendicular to the above axis.

d. The elevation and cross level vial bubbles shall be centered within a half of the width of the graduation line.

e. The coarse azimuth scale index indicating 3200 mils within a half of the width of the graduation line.

f. The coarse elevation scale index indicating 800 mils within a half of the width of the graduation line.

g. The indices for the graduated scales on the azimuth and elevation knobs indicating zero within 0.5 mil.

3.7 performance.

3.7.1 Elevation travel. Rotation of the elevation knob shall cause the line of sight to depress a minimum of 800 mils and elevate a minimum of 100 mils from the position specified in 3.6.

3.7.2 plumb travel. The line of sight shall track a plumb line within 1.0 mil when elevated and depressed through the range specified in 3.7.1 and the position specified in 3.6. The total lateral spread shall not exceed 1.0 mil.

3.7.3 Lateral displacement. The line of sight shall not exceed 0.5 mil lateral displacement when the elevation motion is reversed by means of the elevation knob.

3.7.4 Vertical displacement. The line of sight shall not exceed 0.5 mil vertical displacement when the azimuth motion is reversed by means of the elevation knob.

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3.7.5 Horizontal travel. The line of sight shall not vary in elevation in excess of 1.0 mil (total spread of 2 mils) when rotated 6400 mils in azimuth by means of the azimuth knob.

3.7.6 Angular accuracy.

3.7.6.1 Elevation- the angular accuracy with backlash included shall be within 2 mils over the range of elevation travel.

3.7.6.2 Azimuth. The angular accuracy with backlash included shall be within 2 mils over the range of horizontal travel.

3.7.7 Backlash.

3.7.7.1 Elevation. When the amount of backlash at each elevation reading, taken in accordance with 3.7.6.1, is added to the elevation error at that setting, the sum shall not exceed 2.0 mil.

3.7.7.2 Azimuth. When the amount of backlash at each azimuth reading, taken in accordance with 3.7.6.2, is added to the azimuth error at that setting, the sum shall not exceed 2.0 mil.

3.8 Operability.

3.8.1 Telescope clamping mechanism. The telescope clamping mechanism shall be operable in the unlocked and locked positions when using a mandrel test piece having a diameter of .624 - .001 inches. The mandrel shall be fabricated of aluminum alloy 2024 or 6061, T6, QQ-A-225/6 or /8; or ASTM B211. Surface finish 63 minimum. The mandrel shall be anodized. Dimensions apply after anodizing. These requirements shall be met at the ambient temperature and while the mount is subjected to the operating temperatures of 3.3.3.1.

3.8.1.1 Unlocked position. The mandrel test piece shall rotate freely within its mating diameter and the clamping mechanism shall operate freely about its axis in a fore and aft direction.

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3.8.1.2 Locked position. When locked in place by applying a force of 10 to 12 in-lbs to the wing nut and 15 to 17 in-lbs to the locking lever, there shall be no rotation of the mandrel within the inside diameter of the clamp when a 35 in-lb torque is applied. When locked in place with above forces, the clamping mechanism shall not rotate about its axis when a 15 in-lb torque is applied.

3.8.2 Mount locking latch. The mount locking latch shall be operable at the ambient temperature of 4.2.1 and while subjected to the operating temperatures of 3.3.3.1.

3.8.3 Locking knobs. Subsequent to rotating the elevation and azimuth locking knobs to a locked position with a 15 inch-pound torque, there shall be no movement in their respective mechanisms when an 8 pound-inch torque is applied to the elevation and azimuth knobs. This requirement shall be met at the ambient temperature and while the mount is subjected to the operating temperatures of 3.3.3.1.

3.8.4 Azimuth slip scales The micrometer and coarse azimuth slip scales shall be operable at the ambient temperature and while subjected to the temperature requirements of 3.3.3.1.

3.8.4.1 Torque, coarse azimuth scale. The coarse azimuth scale shall withstand a torque of 3.o inch-pounds without slippage in both the clockwise and counter-clockwise directions at ambient temperature.

3.8.5 Torque. The ruining torque in inch-pounds on all knobs shall be within the limits specified in Table I.

TABLE I. Torque values (inch-pounds)

Knob	+60°F to 90°F (+16°C to +32°C) Standard Temperature Min	Max	-50°F and +145°F (-46°C and +63°C) Extreme Operating Temperature Min	Max
Elevation	0.5	4.0	0.5	9.0
Azimuth	0.5	4.0	0.5	9.0

3.9 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

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3.10 Workmanship. The workmanship requirements of MIL-F-13926 shall be met. The exterior protective surface finishes shall be free of any defects which affect their function, protective qualities or appearances.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-F-13926 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Special sampling inspections (see 4.5).

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4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a. Temperature +50°F (+16°C) to +90°F (+32°C)

b. Relative humidity 90% RH max.

c. Pressure 28 to 32 inches of Hg

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer consisting of three (3) each of the following items, and one each of all remaining items covered by QAPs in TDPL for inspection in accordance with provisions of 4.3.2.

Name	Drawing	QAP	Quantity
Mount, Telescope	9360169		3
Knob Assembly	9360178	9360178	3
Scale Assembly	9360180	9360180	3
Knob Assembly	9360182	9360182	3
Holder Assembly	9360184	9360184	3
Shaft Assembly	11741740	11741740	3
Shaft Assembly	11741741	11741741	3
Housing	11741770	11741770	3
Gear	11741774	11741774	3
Bracket Assembly	11741792	11741792	

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term “inspection lot” is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following classification of characteristics paragraphs. The contractor’s quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table II below, using the inspection levels stated in the Classification of Characteristics paragraphs.

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TABLE II. Attributes sampling inspection.

Lot Size	Inspection Levels I	II	III	IV
2 to 8	*	*	*	*
9 to 15	*	*	*	13
16 to 25	*	*	*	13
26 to 50	*	*	32	13
51 to 90	*	*	32	13
91 to 150	*	125	32	13
151 to 280	*	125	32	32
281 to 500	*	125	32	32
501 to 1200	*	125	80	50

Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the elvel of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the

contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with DLAM 4145.8 and Section 5

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herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table III shall be performed on a l00% basis.

TABLE III. Function III. Functional percent inspection.

Characteristic	Requirement	Test Procedure
Critical:	None	N/A
Special:		
11. Coarse Azimuth Scale Torque	3.8.4.1	4.7.8.6
Major:		
101. Tritium contamination	3.4	4.7.4
102. Illumination	3.5	4.7.5
103. Performance	3.7	4.7.6 and 4.7.7
104. Operability	3.8	4.7.8
105. Workmanship	3.10	4.7.9
Minor		

4.5 Special sampling Three mounts shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall have met the requirements and tests of Table IV and shall then meet the tests in Table 111. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE IV. Special sampling.

Characteristic	Requirement	Test Procedure
Critical:	None	N/A
Minor:		
201 l Vibration	3.3.1	4.7.1
202. Shock	3.3.2	4.7.2
203. Temperature	3.3.3	4.7.3

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Minor:	None	N/A

4.5.1 Failure sample. should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Expection equipment.

4.6.1 Government Furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government sign. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor desire. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table V (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

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TABLE V. Test equipment

Item	Name/Description	Drawings Number
1.	Elevation and Cross Leveling Fixture	6523553
2.	Azimuth Test Fixture	7691596
3.	Adapter, Final Inspection, Elevation	11750708
4.	Adapter, Final Inspection, Azimuth	11750709
5.	Alignment Gage, Final Inspection, Azimuth	11750712
6.	Inspection Aid	8658940
7.	Dioptometer	7680631
8.	Mounting Bracket for Dioptometer	See 4.6.2.3.1
9.	Wall Target	See 4.6.2.3.2

4.6.2.3.1 Mounting bracket for dioptometer. The mounting bracket shall consist of a sleeve and mandrel joined by a connecting arm. The geometric axis of the sleeve and mandrel shall lay in the same vertical plane within .001 inch. The geometric axes of the sleeve and mandrel shall also be horizontal and parallel to each other with .001 inch. The geometric axis of the sleeve shall be l.50 + .25 inch above the geometric axis of the mandrel. The inside diameter of the sleeve shall be capable of accepting the dioptometer, 7680631 (outside diameter 1.18 -.01 inch). The sleeve shall secure and maintain the dioptometer in a horizontal position. The mandrel shall have a length of 1.000 + .250 inch and an outside diameter of .624 - .001 inch. Install the dioptometer in the sleeve of the bracket. Rotate the dioptometer until its vertical reticle line lays in the same vertical plane that contains the geometric axes of the sleeve and mandrel. Lock the dioptometer in this position.

4.6.2.3.2 Wall target. A calibrated wall target shall be utilized in measuring the elevation accuracy requirements of the mount. The wall target shall contain a vertical and horizontal crossline pattern together with limiting displacement marks needed to determine line of sight deviations (azimuth shift) during elevation travels. This wall target shall be calibrated for a measured distance between 6.0 and 9.0 feet, as measured from the mechanical rotational axis (elevation) of the mount.

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4.7 Methods of inspection.

4.7.1 Vibration. Conduct the vibration test with the mount positioned on an adapter affixed to a standard vibrating machine that is capable of providing the amplitude and frequency specified in 3.3.1. The adapter shall position the mount as specified in 3.3.1 during the vibration cycle. Unless otherwise specified, the test equipment shall be in accordance with the "Test Facilities" requirements of MIL-STD-810. The applied motion shall be in a vertical plane for the time period specified. Upon completion of the vibration cycle, the mount shall be removed from the vibration adapter and subjected to the test in 4.7.4 through 4.7.8. If mount does not comply with any requirements of 3.4 through 3.8, the mount shall be rejected.

4.7.2 Shock. Conduct the shock test with the mount positioned on an adapter affixed to a shock test machine. The adapter shall be capable of holding and positioning the mount as specified in 3.3.2. The number of shocks and applied directions shall be as specified in 3.3.2. The shock pulse shape, peak values, and normal time durations for the positions indicated shall be a specified in 3.3.2. Position the mount (scale and knobs) as specified prior to shock testing. Subsequent to the shock applications (18), the mount shall be removed from the shock test adapter and subjected to the tests in 4.7.4 through 4.7.8. If mount does not comply with any requirements of 3.4 through 3.8, the mount shall be rejected.

4.7.3 Storage/operating temperature The testing equipment utilized in this test shall conform to the requirements of MIL-STD-810. The mount shall be placed in the test chamber and the temperature of the chamber reduced, over a period of 4 hours plus or minus 15 minutes, from room ambient temperature. The mount shall be maintained at this temperature for a minimum of 4 hours. Upon completion of the 4 hour exposure, and while at -50 degrees F (-46 degrees C), the mount shall be subjected to the operability tests in 4.7.8.1 through 4.7.8.6. Upon completion of the low temperature operability tests, the temperature Of the chamber shall be raised, over a period of 4 hours plus or minus 15 minutes, to room ambient temperature and the mount allowed to stabilize at this temperature for 2 hours plus or minus 15 minutes at which time the mount will be subjected to the test of 4.7.8.5. Upon completion of the room ambient temperature test, the temperature of the chamber shall be raised, over a period of 4 hours plus or minus 15 minutes, to +160 degrees F (+71 degrees C). The mount shall be maintained at this temperature for a minimum of 4 hours. Upon completion of the 4 hour exposure, the temperature of the chamber shall be reduced, over a period of 30 minutes plus of minus 2 minutes, to +145 degrees F (+63 degrees

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c). The mount shall be maintained at this extreme temperature for a minimum of 2 hours. Upon completion of the 2 hour exposure and while at +145 degrees F (+63 degrees C), the mount shall again be subjected to the operability tests in 4.7.8.1 through 4.7.8.6. Upon completion of the high temperature operability tests, the temperature of the chamber shall be reduced, over a period of 4 hours plus or minus 15 minutes, to room ambient temperature. The mount shall then be removed from the test chamber and subjected to the test in 4.7.4 and then a visual examination for illumination of light sources. It shall meet the requirements of 3.4 and 3.5., respectively, subsequent to the temperature exposures of 3.3.3.1. The mount shall have conformed to the requirements of 3.3.3.1 at each temperature extreme.

NOTE: When chamber temperatures are being raised or reduced, the rate of temperature change shall not exceed +2°F (1.1°C) per minute.

4.7.4 External tritium contamination. The contamination test shall be performed by wiping the item under test with filter paper moistened in distilled water. The damp filter paper shall be placed in a container with the proper percentage of scintillation liquid, which is determined by the type of scintillation test equipment used to determine the actual contamination. Insert the sample into this system, count a minimum of 2000 disintegrations and convert the readout into microcuries (222 DPM 10 = 1 microcurie). The measured contamination shall be within the limits specified in 3.4. The liquid scintillating system must be calibrated on six-month intervals, to a standard traceable to an N.I.S.T. standard.

4.7.5 Illumination. The illumination test shall be performed under simulated conditions specified in 3.5. The luminosity test shall be performed in a dark room having a variably controlled light source. To perform this test, the observer must be dark adapted by being in a darkened room for a minimum of 15 minutes before viewing scale markings, indices, level vial bubbles, and markings. All scale markings, indices, level vial bubbles, and markings shall be clearly distinguishable when viewed, at a distance of 18 to 20 inches from the viewer's eye, with and without a 50% neutral density filter while the light level is varied gradually from zero to ten foot-candles over a period of one to two minutes.

4.7.6 Orientation, elevation accuracy. The testing equipment specified in Table IV shall be utilized in performing the elevation travel and accuracy tests. The elevation and cross leveling fixture shall be positioned on a vibration free surface and cross leveled in the horizontal plane by means of a certified

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10 second level. The Adapter, 11750708, shall be affixed to the elevation and cross leveling fixture and cross leveled by means of the 10 second level so that the interfacing (female) dovetail meets the requirements of 3.6a and 3.6b. position the mount on the elevation adapter. Orient the mount as specified in 3.6d through 3.6g. Adjust the dioptometer eyepiece for best focus of the dioptometer reticle. Insert the mandrel portion of the mounting bracket, with dioptometer, into the telescope clamping mechanism of the mounting bracket, with dioptometer, into the telescope clamping mechanism of the mount. The eyepiece of the dioptometer shall be on the side.of the mount that contains the elevation knob. Rotate the mounting bracket, in the clamping mechanism, until the dioptometer vertical reticle line is plumb. Lock the clamping mechanism. Position the wall target at its calibrated distance, focus the dioptometer on the wall target by means of the objective assembly. Elevate the target until the intersecting point of the horizontal and vertical target line is coincident to the intersecting point of the dioptormeter horizontal and vertical reticle lines. Plumb the vertical target line independent of the dioptometer vertical reticle line.

NOTE : The condition of 3.6d shall be established with backlash removed from the elevation mechanism for rotation in depression.

4.7.6.1 Elevation travel With backlash removed for direction of travel, and while directing a line of sight through the center of the dioptometer reticle pattern to the vertical target line, depress the line of sight of the mount in 200 mil increments by means of its elevation knob. Return the line of sight through the dioptometer, to the horizontal plane by elevating the elevation and cross leveling fixture. Repeat the above steps until the mount is at maximum depression. Position the inspection aid (8658940) on the gunner's quadrant seats provided on the elevation adapter and verify the mil by measuring the angle established on the elevation adapter. Return the mount and elevation and cross leveling fixture to zero mil elevation. By means of the elevation knob, elevate the line of sight of the mount to its maximum elevation position. Again, by means of the Inspection Aid (8658940) verify the mil value indicated on the elevation scale and micrometer knob of the mount. The mount shall conform to the requirements of 3.7.1.

4.7.6.2 plumb travel. Perform this test concurrent with the test in 4.7.6.1. During the excursions in depression and elevation, the vertical reticle line of the dioptometer, at the

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center of the reticle pattern, shall track the vertical target line within the tolerance specified in 3.7.2, as measured on the target. Caution shall be exercised to assure that this test is conducted with backlash removed for direction of travel.

4.7.6.3 Lateral displacement perform this test in sequence to the test in 4.7.6.1 and 4.7.6.2. with backlash removed for direction of travel and the dioptometer reticle pattern superimposed on the vertical target-line as specified in 4.7.7.1, track the vertical target line for approximately 50 mils by means of the elevation knob of the mount. Then, with backlash included, reverse direction of the elevation knob for approximately 50 mils. Any lateral displacement of the dioptometer vertical reticle line, as a result of this operation shall not exceed the tolerance specified in 3.7.3 as measured on the target. Tests for lateral displacement shall be performed within 50 mils of each extremity of the elevation scale. A third test point for lateral displacement shall be performed at approximately mid-range of the elevation scale. All points of tests for lateral displacement shall be randomly selected within the above specified areas.

4.7.6.4 Elevation angular accuracy Perform this test at 750 mils, 1000 mils, 1200 mils, 1400 mils, and 1550 mils. With backlash removed from fixture rotation in elevation, elevate the adapter to zero mil in elevation. The zero mil elevation position of the elevation adapter shall be verified with the Inspection Aid, 8658940. With backlash removed for rotation of a line sight in depression, verify that the dioptometer reticle lines are superimposed on the target lines, and that the mount conforms to the orientation requirements of 3.6d through 3.6g. Depress the line of sight of the mount, by means of its elevation knob, to the first randomly selected elevation setting. Then, by means of the elevation mechanism of the test fixture, return the line of sight of the mount to the horizontal plane until the dioptometer reticle lines are again superimposed on the target lines without overtravel. Utilizing the inspection aid, measure the elevation angle on the adapter. The measured angle recorded on the inspection aid must equal the angle indicated on the scale and micrometer of the mount within the tolerance specified in 3.7.6.1. Utilizing the foregoing method, proceed to inspect the elevation accuracy of the mount at the three remaining selected elevation settings. The one elevation setting below 800 mils shall be inspected from a zero mil setting of the fixture and a 800 mil setting of the mount. Record the elevation error for each elevation setting tested for use in the elevation backlash test. Caution must be exercised throughout all elevation accuracy measurements to exclude backlash from the mount and test fixture for their appropriate directions of travel. The

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elevation accuracy of the mount shall conform to the requirements of 3.7.6.1 and meet the requirements of 3.6d.

4.7.6.5 Elevation backlash. Perform this test in sequence to the test in 4.7.6.4. At each of the selected elevation settings, record the elevation (mil) value when the reticle lines are coincident with the target lines. With backlash excluded for direction of travel, override the target line (horizontal) by approximately 50 mils utilizing the elevation knob of the mount. Then, with backlash included, reverse direction of the elevation knob of the mount. Then, with backlash included, reverse direction of the elevation knob and reestablish coincidence of the reticle and target lines (horizontal), without Overtravele Again record the elevation (mil) value as read on the scale and micrometer knob. The difference between the first and second recorded (mil) value is the amount of backlash. At the selected elevation setting, the amount of backlash for that elevation position shall not exceed the tolerance specified in 3.7.7.1. Perform the elevation backlash test at each of the selected elevation settings tested in 4.7.6.4.

4.7.7 Orientation. azimuth accuracy. The testing equipment specified in Table IV shall be utilized in performing the azimuth accuracy tests. The azimuth test fixture shall be positioned on a vibration free surface and cross leveled in the horizontal plane by means of a certified 10 second level. Rotate the wheel assembly of the fixture to the zero azimuth setting. Lock the assembly in this position. Verify that the mechanical axis of the target projector collimator is aligned to the pivotal axis of the wheel assembly of the fixture. The height of the collimator shall be such that its optical axis is coincident to the axis of the sleeve of the mounting bracket when the mount, with mounting bracket and dioptometer affixed, is inserted in the Adapter, 11750709. With the Adapter, 11750709, mounted on the wheel assembly, the Alignment Gage, 11750712, together with the dioptometer, shall align the adapter so that its interfacing (female) dovetail meets the requirement of 3.6a and 3.6b. The alignment gage, with dioptometer, shall also be utilized to establish the mounting dovetail surface of the mount parallel to the optical axis of the collimator. Position the mount on the azimuth adapter. Adjust and position the dioptometer and mounting bracket on the mount as specified in 4.7.6. Orient the mount as specified in 3.6.d through 3.6g. Verify that the vertical and horizontal reticle lines of the dioptometer and collimeter target lines are superimposed. Then ascertain that the optical axis of the dioptometer is horizontal by rotating the wheel assembly of the fixture to 1600 mils, then counter-rotating the mount, by means of the azimuth knob, to return the dioptometer vertical reticle line coincident to the collimator

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vertical target line. Correct any elevation error, as evidence by the lack of coincidence of the horizontal reticle and collimator target lines, by adjusting the clamping mechanism for elevation of the line of sight. Then rotate the wheel assembly of the fixture to 4800 mils and repeat the above procedure to level the dioptometer. Repeat the entire procedure until the dioptometer is horizontal at 1600 mils, zero mils, and 4800 mils without interim adjustments. Return the fixture to the zero mil azimuth setting and the dioptometer reticle coincident to the target collimator vertical and horizontal lines.

4.7.7.1 Horizontal travel. Verify that the intersecting point of the dioptometer reticle lines is coincident with the horizontal target line. With backlash removed for direction of travel, rotate the line of sight of the mount through 6400 mils, in precise 800 mil increments by means of the azimuth knob. At each 800 mil incremental position of the mount, counter-rotate the wheel assembly of the fixture 800 mils to reestablish coincidence of the dioptometer verticle line and vertical target line. At each incremental position,note any lack of coincidence of the horizontal reticle line and horizontal target line. any vertical displacement of the line of sight through the intersecting point of the dioptometer reticle lines, with reference to the horizontal target line, shall not exceed the tolerance specified in 3.7.5 as measured on the target collimator and meet the requirements of para 3.6d.

NOTE: Care shall be exercised to assure that backlash is excluded in all horizontal travel measurements. The 800 mil incremental positions of the mount and test fixture shall be accomplished without overtravel.

4.7.7.2 Vertical displacement This test may be conducted concurrent with the test in 4.7.7.1. At each 800mil incremental position, reverse the azimuth motion of the mount by means of the azimuth knob a minimum of 50 mils. As a result of this reversing operation, the line of sight, through the intersection of the dioptometer reticle lines, shall not displace vertically in excess of the tolerance specified in 3.7.4, as measured on the target collimator.

4.7.7.3 Azimuth angular accuracy. Perform this test at 800 mil incremental position (8) as the line of sight of the mount is rotated 6400 mils in azimuth. Verify that the fixture wheel assembly is positioned at zero mil azimuth with backlash removed for appropriate movement in azimuth. With backlash removed for rotation of the line of sight of the mount to 800 mils, verify that the dioptometer vertical reticle line is superimposed on the vertical target line, and that the mount conforms to the

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requirements of 3.6d through 3.6g, Rotate the line of sight of the mount to 800 mils without overtravel. Counter-rotate the fixture wheel assembly and again obtain coincidence of the dioptometer vertical reticle line and vertical target line without overtravel. my difference between the mil value recorded on the mount and the scale of the fixture is azimuth accuracy error. Record any azimuth accuracy error for use in the azimuth backlash test. Utilizing the foregoing method, proceed to inspect the azimuth accuracy of the mount at the seven (7) remaining incremental positions. Record the azimuth error for each of the eight (8) incremental positions tested. Caution must be exercised throughout all azimuth accuracy measurements to exclude backlash from the mount and test fixture for their appropriate direction of travel. The azimuth accuracy of the mount shall conform to the requirements of 3.7.6.2 and meet the requirements of para 3.6d.

4.7.7.4 Azimuth backlash Perform this test in sequence to the test in 4.7.7.3, at each of the eight (8) incremental positions. Subsequent to the azimuth angular accuracy measurements rotate the mount, by means of the azimuth knob, to an azimuth setting of approximately 850 mils. Then, with backlash included, reverse direction of the knob and again obtain coincidence of the dioptometer vertical reticle line and vertical target line without overtravel. Record the azimuth mil value as read on the azimuth knob. any azimuth change from the original 800 mil setting, as read on the azimuth knob scale, is the amount of backlash. At the selected azimuth position, the amount of backlash for that position added to the azimuth accuracy error recorded for that position, shall not exceed the tolerance specified in 3.7.2.

4.7.8 Operability. The following test shall be conducted with the mount positioned on a vibration free surface that has a mating dovetail that maintains the mount as specified in 3.6a and 3.6b, and a mandrel that meets the physical requirements specified in 3.8.1. The torque wrench type tester for testing knobs and clamping mechanism torque shall have a range of zero to fifty inch-pounds. The torque wrench type tester for testing rotation of the mandrel shall have a range of zero to fifty inch-pounds. The accuracy of the torque wrench type testers shall be five percent throughout its entire zone of excursion. The testing equipment shall be equipped with Government approved adapters to accommodate testing of the appropriate assembly.

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4.7.8.1 Telescope Clamping mechanism. unlocked Position. When subjected to a tactile examination, the function of the locking lever shall meet the requirements of 3.8.1.1. When the mandrel is inserted into the telescope holder, it shall rotate freely for 360 degrees and the telescope holder move fore and aft freely.

4.7.8.2 Telescope clamping mechanism. locked position Insert the mandrel into the telescope holder. Apply a force of 15 to 17 inch-pounds to the locking lever and 10 to 12 inch-pounds to the wing nut. Apply the specified torques to the mandrel and telescope holder, in turn. The Telescope clamping mechanism shall meet the reqirements of 3.8.1.2.

4.7.8.3 Mount locking latch Install and remove the mount from the dovetail on the stand a minimum of three (3) times (3 installations). The mount shall meet the requirements of 3.8.2, at the temperature specified.

4.7.8.4 Locking knobs. Rotate the azimuth and elevation locking knobs to a locked position with the appropriate torque wrench and adapter. The azimuth and elevation mechanisms shall meet the requirements of 3.8.3 when the specified torque is applied to the applicable knob, at the temperature indicated.

4.7.8.5 Azimuth slip scales. When subjected to a tactile examination for operability, the micrometer knob slip scales (azimuth and elevation) and azimuth scale shall meet the requirements of 3.8.4, at the temperature indicated.

4.7.8.6 Torque, coarse azimuth This test shall be performed on complete M64A1 sight Mount Assembly (P/N 9360169) units. The sight assembly shall be mounted via the dovetail such that it is possible to level the bubble vials using the azimuth and elevation knobs and with sufficient security to assure that the level will not change during the test. The torque shall be applied in a uniformly increasing reamer at a rate not to exceed 1.0 inch-pound per second and the actual torque value at which the scale initially slips shall be recorded. The test equipment shall not exert a force on the coarse azimuth scale in a downward direction greater than 1.1 pounds while the torque is being measured. Prior to performing the test, the azimuth control knob shall be used to rotate the coarse azimuth scale at least 400 mils in the clockwise direction stopping with the 0 mil mark on the scale aligned with the index arrow. The counterclockwise torque value should now be obtained. The azimuth control knob should then be used to rotate the coarse azimuth scale 400 mils in the counterclockwise direction and the clockwise torque value should be obtained. The coarse azimuth scale shall then be

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rotated clockwise approximately 180 degrees with the azimuth control knob, stopping with the 3200 mil mark on the index. Counterclockwise and clockwise torque values shall be obtained at the 3200 mil position in the same manner as that detailed for the 0 mil position. Extreme care must be taken in the design and operation of the test equipment to assure that not even the slightest movement be imparted to the scale in the same direction, as the torque is being measured, as this could result in erroneous readings. If all four torque values do not equal or exceed the minimum value specified in paragraph 3.8.4.1, the sightunit shall be rejected.

4.7.8.7 Torque. azimuth, and elevation knobs. Utilizing the appropriate torque wrench with adapter, the running torque required to rotate the knobs throughout their range of rotation shall be within the values specified in 3.8.5, for the temperature indicated.

4.7.9 Workmanship. The mount shall be subjected to a visual and tactile examination for evidence of faulty workmanship. The mount shall conform to the requirements of 3.10.

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Special Packaging Instruction PS 9360169 and DLAM 4145.8. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Mount, Telescope, 9360169, is a component of Sightunit, M64A1. The mount introduces measured azimuth and elevation readings to Elbow Telescope, 9360257. The sightunit is used with the U.S. Army mortar systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1).

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c. Requirements for submission of first article sample.

d. Applicable stock number.

e. Packaging requirements, if other than specified in Section 5.

f. Serialization requirements, if applicable.

g. Certificate of conformance for each lot or shipment of product.

6.3 Submission by the contractor.

a) Inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAF-1, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the. Contract Data Requirements List, DD Form 1423 in the contract.

b) Alternative quality conformante provisions for approval. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation to the technical activity responsible for the preparation of this specification.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Special defect. A defect, other than Critical, that judgement and experience indicate may, depending upon the degree of variance from the design requirement:

a. Result in hazardous or unsafe conditions for individuals using, maintaining or depending upon the product, or

b. Prevent performance of the tactical function of major end item.

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6.6 Subject term (key work) listing.

Fire control system
Mortar
Sightunit
Tritium

6.7 Changes from previous issue Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1240 -AB44)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-M-70453A	2. DOCUMENT DATE (YYMMDD)
DOCUMENT TITLE		

NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

REASON FOR RECOMMENDATION

5. SUBMITTER		
a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON (*If applicable*)	e. DATE SUBMITTED (*YYMMDD*)

6. PREPARING ACTIVITY	
a. NAME	b. TELEPHONE (*Include Area Code*)
c. ADDRESS (*Include Zip Code*)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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